

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.			
20							***** MODULAR VALVES *****			816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.			
30										817	USE ONLY FOR TIE IN			
30	U04		MODULAR DBB	K15AL10	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.			
40			MODULAR DBB	K15AL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
50		VAW, 817	MODULAR DBB	K15CL14	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22			
60	U04	VAW	MODULAR DBB	M15AL10	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA1	TO BE USED WITH RAISED FACE FLANGES ONLY.			
70	U04	VAW	MODULAR DBB	M15AL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	BA5	TO BE USED WITH COMPACT FLANGES REQUIRING COMPACT NUTS ONLY. FOR FGP ONLY.			
80		VAW	MODULAR DBB	M15EL10	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.			
90	U04	VAW	MODULAR DBB	M15EL10	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	1.5	2	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117			
100		VAW, 817	MODULAR DBB	M15ZL10	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.			
110	U04	VAW, 817	MODULAR DBB	M15ZL10	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	UDA	FOR USE ON MODULAR VALVES WITH "HARD PIPED" VENT CONNECTIONS.			
120		VAW, 817		K15ZL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# CF X RF INC 625 INC 625 TRIM METAL LO REDUCING - TYPE 2BV	3	3	UDE	USE FOR CONECTION TO EXTRUDED BRANCH FITTINGS ONLY			
130	U04	VAW, 817	MODULAR DBB	K15ML14	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF X RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS			
140	U04	VAW, 817	MODULAR DBB	K15ML14L	3	3	MODULAR VALVE DB&B BALL TYPE 1500# CF X RF INC 625 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
150	U04	VAW, 817	MODULAR DBB	M15ZL10L	1	1	MODULAR VALVE DB&B BALL TYPE 1500# CF x RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	3	3	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
160	U04	VAW	MODULAR DBB	M15EL10L	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	1.5	2	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
170	U04	VAW	MODULAR DBB	M15EL14	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - TYPE R2BVR	3	3	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
180	U04	VAW	MODULAR DBB	M15EL14L	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF INC 625 INC 625 TRIM METAL LO REDUCING - TYPE R2BVR	3	3					
190		VB2					***** BALL VALVES *****							
200														
200			BALL	A15AL13	2	2	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT RB LO	0	0					
210	U04	817	BALL	A15CL13	3	3	BALL 1500# CF INCONEL 625 TRIM AND BODY TRUNNION SOFT RB LO	0	0					
220	U04	817	BALL	A15CH15	4	24	BALL 1500# CF F22 INCONEL 625 TRIM TRUNNION SOFT RB GO	0	0					
230			BALL	B15AL09	0.75	1.5	BALL 1500# RF INCONEL 625 BODY AND TRIM FLOATING SOFT FB LO	0	0					
240		VA5	BALL	B15AL13	2	3	BALL 1500# RF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB LO	0	0					
250	U04	817	BALL	B15CL15	3	4	BALL 1500# CF INCONEL 625 BODY AND TRIM TRUNNION SOFT FB GO	0	0					
260	U04	817	BALL	B15CH15	4	24	BALL 1500# CF F22 INCONEL 625 TRIM TRUNNION SOFT FB GO	0	0					
270							***** THROUGH CONDUIT GATE VALVES *****							
270														
280	U04	817	GATE	W15CH70	3	4	GATE 1500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB HW	0	0					
290	U04	817	GATE	W15CH71	6	24	GATE 1500# CF F22 TRIM INC 625 TRIM THROUGH COND SOLID SLAB GO	0	0					
300							***** GLOBE VALVES *****							
300														
310			GLOBE	L15AL43	0.75	2	GLOBE 1500# RF INC 625 B564 INC 625 TRIM OS&Y HW	0	0					
320	U04	817	GLOBE	L15CL43	3	3	GLOBE 1500# CF INC 625 B564 INC 625 TRIM OS&Y HW	0	0					
330	U04	817	GLOBE	L15CH44	4	6	GLOBE 1500# CF F22M INC 625 TRIM OS&Y GO	0	0					
340							***** CHECK VALVES *****							
340														
350		VC1	CHECK	C15AL51	0.75	1.5	CHECK PISTON 1500# RF INC 625 TRIM 625	0	0					
360	U04		CHECK	C15AL46	2	2	CHECK DUAL WAFER LUG 1500# RF INC 625 TRIM 625 THROU BOLTED	0	0					
370	U04	817	CHECK	C15CL53	3	3	CHECK DUAL DBL FLG 1500# CF INC 625 TRIM 625	0	0					
380	U04	817	CHECK	C15CH53	4	24	CHECK DUAL DBL FLG 1500# CF F22 TRIM 625	0	0					
390							***** PIPE *****							
390														
400														
400		812	PIPE	5358511	0.75	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0					
410			PIPE	5360053	2	2	PIPE WT 7.0MM SMLS 2-1/4CR F22M BE	0	0					
420			PIPE	5360054	3	3	PIPE WT 8.5MM SMLS 2-1/4CR F22M BE	0	0					
430			PIPE	5360055	4	4	PIPE WT 10.0MM SMLS 2-1/4CR F22M BE	0	0					
440			PIPE	5360056	6	6	PIPE WT 13.0MM SMLS 2-1/4CR F22M BE	0	0					
450			PIPE	5360057	8	8	PIPE WT 16.0MM SMLS 2-1/4CR F22M BE	0	0					
460			PIPE	5360059	10	10	PIPE WT 19.0MM SMLS 2-1/4CR F22M BE	0	0					
470			PIPE	5360060	12	12	PIPE WT 22.0MM SMLS 2-1/4CR F22M BE	0	0					
480			PIPE	5360072	14	14	PIPE WT 24.0MM SMLS 2-1/4CR F22M BE	0	0					
490			PIPE	5360073	16	16	PIPE WT 27.0MM SMLS 2-1/4CR F22M BE	0	0					
			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
			ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE		CORR. ALLOW. 3 MM		SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -40 / 155 °C		12/16/2024	BY OZAC	U04	Page 1 of 5	1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
500			PIPE	5360076	18	18	PIPE WT 30.0MM SMLS 2-1/4CR F22M BE	0	0	
510			PIPE	5360077	20	20	PIPE WT 33.0MM SMLS 2-1/4CR F22M BE	0	0	
520			PIPE	5360052	24	24	PIPE WT 38.5MM SMLS 2-1/4CR F22M BE	0	0	
530			PIPE	5360093	30	30	PIPE WT 47.5MM SMLS 2-1/4CR F22M BE	0	0	
540										
540							***** FITTINGS *****			
550										
550			ELBOW	5392870	0.75	1.5	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0	
560			ELBOW	5406638	2	2	ELL 90 DEG LR WT 7.0MM 2-1/4CR F22M	0	0	
570			ELBOW	5406639	3	3	ELL 90 DEG LR WT 8.5MM 2-1/4CR F22M	0	0	
580			ELBOW	5406640	4	4	ELL 90 DEG LR WT 10.0MM 2-1/4CR F22M	0	0	
590			ELBOW	5406641	6	6	ELL 90 DEG LR WT 13.0MM 2-1/4CR F22M	0	0	
600			ELBOW	5406642	8	8	ELL 90 DEG LR WT 16.0MM 2-1/4CR F22M	0	0	
610			ELBOW	5406643	10	10	ELL 90 DEG LR WT 19.0MM 2-1/4CR F22M	0	0	
620			ELBOW	5406644	12	12	ELL 90 DEG LR WT 22.0MM 2-1/4CR F22M	0	0	
630			ELBOW	5406602	14	14	ELL 90 DEG LR WT 24.0MM 2-1/4CR F22M	0	0	
640			ELBOW	5406603	16	16	ELL 90 DEG LR WT 27.0MM 2-1/4CR F22M	0	0	
650			ELBOW	5406606	18	18	ELL 90 DEG LR WT 30.0MM 2-1/4CR F22M	0	0	
660			ELBOW	5406607	20	20	ELL 90 DEG LR WT 33.0MM 2-1/4CR F22M	0	0	
670			ELBOW	5406610	24	24	ELL 90 DEG LR WT 38.5MM 2-1/4CR F22M	0	0	
680			ELBOW	5406633	30	30	ELL 90 DEG LR WT 47.5MM 2-1/4CR F22M	0	0	
690			ELBOW	5392876	0.75	1.5	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0	
700			ELBOW	5407031	2	2	ELL 45 DEG LR WT 7.0MM 2-1/4CR F22M	0	0	
710			ELBOW	5407032	3	3	ELL 45 DEG LR WT 8.5MM 2-1/4CR F22M	0	0	
720			ELBOW	5407033	4	4	ELL 45 DEG LR WT 10.0MM 2-1/4CR F22M	0	0	
730			ELBOW	5407034	6	6	ELL 45 DEG LR WT 13.0MM 2-1/4CR F22M	0	0	
740			ELBOW	5407035	8	8	ELL 45 DEG LR WT 16.0MM 2-1/4CR F22M	0	0	
750			ELBOW	5407036	10	10	ELL 45 DEG LR WT 19.0MM 2-1/4CR F22M	0	0	
760			ELBOW	5407037	12	12	ELL 45 DEG LR WT 22.0MM 2-1/4CR F22M	0	0	
770	U04		ELBOW	5407062	14	14	ELL 45 DEG LR WT 24.0MM 2-1/4CR F22M	0	0	
780			ELBOW	5407063	16	16	ELL 45 DEG LR WT 27.0MM 2-1/4CR F22M	0	0	
790			ELBOW	5407066	18	18	ELL 45 DEG LR WT 30.0MM 2-1/4CR F22M	0	0	
800			ELBOW	5407067	20	20	ELL 45 DEG LR WT 33.0MM 2-1/4CR F22M	0	0	
810			ELBOW	5407042	24	24	ELL 45 DEG LR WT 38.5MM 2-1/4CR F22M	0	0	
820			ELBOW	5407096	30	30	ELL 45 DEG LR WT 47.5MM 2-1/4CR F22M	0	0	
830										
830			TEE	5392879	0.75	1.5	TEE SCH XXS LTCS A420 WPL6	0	0	
840			TEE	5407214	2	2	TEE WT 7.0MM 2-1/4CR F22M	0	0	
850			TEE	5407215	3	3	TEE WT 8.5MM 2-1/4CR F22M	0	0	
860			TEE	5407216	4	4	TEE WT 10.0MM 2-1/4CR F22M	0	0	
870			TEE	5407217	6	6	TEE WT 13.0MM 2-1/4CR F22M	0	0	
880			TEE	5407218	8	8	TEE WT 16.0MM 2-1/4CR F22M	0	0	
890			TEE	5407219	10	10	TEE WT 19.0MM 2-1/4CR F22M	0	0	
900			TEE	5407220	12	12	TEE WT 22.0MM 2-1/4CR F22M	0	0	
910			TEE	5407162	14	14	TEE WT 24.0MM 2-1/4CR F22M	0	0	
920			TEE	5407163	16	16	TEE WT 27.0MM 2-1/4CR F22M	0	0	
930			TEE	5407166	18	18	TEE WT 30.0MM 2-1/4CR F22M	0	0	
940			TEE	5407167	20	20	TEE WT 33.0MM 2-1/4CR F22M	0	0	
950			TEE	5407159	24	24	TEE WT 38.5MM 2-1/4CR F22M	0	0	
960			TEE	5407211	30	30	TEE WT 47.5MM 2-1/4CR F22M	0	0	
970			FLANGE	5392877	0.75	1	TEE RED SCH XXS LTCS A420 WPL6	1	1.5	
980			FLANGE	5407451	0.75	1.5	TEE RED BW WT 7.0MM / XXS 2-1/4CR F22M	2	2	
990			FLANGE	5407452	0.75	1.5	TEE RED BW WT 8.5MM / XXS 2-1/4CR F22M	3	3	
1000			FLANGE	5407453	2	2	TEE RED BW WT 8.5MM / 7.0MM 2-1/4CR F22M	3	3	
1010			FLANGE	5407460	2	2	TEE RED BW WT 10.0MM / 7.0MM 2-1/4CR F22M	4	4	
1020			FLANGE	5407454	3	3	TEE RED BW WT 10.0MM / 8.5MM 2-1/4CR F22M	4	4	
1030			FLANGE	5407455	10	10	TEE RED BW WT 22.0MM / 19.0MM 2-1/4CR F22M	12	12	
1040			FLANGE	5407456	10	10	TEE RED BW WT 24.0MM / 19.0MM 2-1/4CR F22M	14	14	
1050			FLANGE	5407457	12	12	TEE RED BW WT 24.0MM / 22.0MM 2-1/4CR F22M	14	14	
1060			FLANGE	5407458	10	10	TEE RED BW WT 27.0MM / 19.0MM 2-1/4CR F22M	16	16	
1070			FLANGE	5407459	12	12	TEE RED BW WT 27.0MM / 22.0MM 2-1/4CR F22M	16	16	
1080			FLANGE	5407386	16	16	TEE RED BW WT 30.0MM / 27.0MM 2-1/4CR F22M	18	18	
1090			FLANGE	5407412	10	10	TEE RED BW WT 33.0MM / 19.0MM 2-1/4CR F22M	20	20	
1100			FLANGE	5407413	12	12	TEE RED BW WT 33.0MM / 22.0MM 2-1/4CR F22M	20	20	
1110	U04		TEE RED	5407384	14	14	TEE RED BW WT 33.0MM / 24.0MM 2-1/4CR F22M	20	20	
1120			FLANGE	5407383	16	16	TEE RED BW WT 33.0MM / 27.0MM 2-1/4CR F22M	20	20	
1130			FLANGE	5407387	18	18	TEE RED BW WT 33.0MM / 30.0MM 2-1/4CR F22M	20	20	
1140			FLANGE	5407393	16	16	TEE RED BW WT 38.5MM / 27.0MM 2-1/4CR F22M	24	24	
1150			FLANGE	5407396	18	18	TEE RED BW WT 38.5MM / 30.0MM 2-1/4CR F22M	24	24	
1160			FLANGE	5407391	20	20	TEE RED BW WT 38.5MM / 33.0MM 2-1/4CR F22M	24	24	
1170			FLANGE	5407380	24	24	TEE RED BW WT 47.5MM / 38.5MM 2-1/4CR F22M	30	30	
1180										
1180	U04		CAP	5392880	0.75	1.5	CAP BW SCH XXS LTCS A420 WPL6	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -40 / 155 °C	12/16/2024	BY OZAC	U04	Page 2 of 5	1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1190			CAP	5407154	2	2	CAP WT 7.0MM 2-1/4CR F22M	0	0	
1200			CAP	5407155	3	3	CAP WT 8.5MM 2-1/4CR F22M	0	0	
1210			CAP	5407156	4	4	CAP WT 10.0MM 2-1/4CR F22M	0	0	
1220			CAP	5407157	6	6	CAP WT 13.0MM 2-1/4CR F22M	0	0	
1230			CAP	5407158	8	8	CAP WT 16.0MM 2-1/4CR F22M	0	0	
1240			CAP	5407185	10	10	CAP WT 19.0MM 2-1/4CR F22M	0	0	
1250			CAP	5407186	12	12	CAP WT 22.0MM 2-1/4CR F22M	0	0	
1260			CAP	5407102	14	14	CAP WT 24.0MM SMLS 2-1/4CR F22M	0	0	
1270			CAP	5407103	16	16	CAP WT 27.0MM 2-1/4CR F22M	0	0	
1280			CAP	5407106	18	18	CAP WT 30.0MM SMLS 2-1/4CR F22M	0	0	
1290			CAP	5407107	20	20	CAP WT 33.0MM SMLS 2-1/4CR F22M	0	0	
1300			CAP	5407129	24	24	CAP WT 38.5MM SMLS 2-1/4CR F22M	0	0	
1310			CAP	5407151	30	30	CAP WT 47.5MM 2-1/4CR F22M	0	0	
1320										
1320			RED CONC	5392881	0.75	1	REDUCER CONC SCH XXS LTCS A420 WPL6	1	1.5	
1330			RED CONC	5407203	0.75	1.5	REDUCER CONC WT 7.0MM/SCH XXS 2-1/4CR F22M	2	2	
1340			RED CONC	5407204	1.5	1.5	REDUCER CONC WT 8.5MM/SCH XXS 2-1/4CR F22M	3	3	
1350			RED CONC	5407205	1.5	1.5	REDUCER CONC WT 10.0MM/SCH XXS 2-1/4CR F22M	4	4	
1360	U04		RED CONC	5407206	2	2	REDUCER CONC WT 8.5MM/7.0MM 2-1/4CR F22M	3	3	
1370			RED CONC	5407237	2	2	REDUCER CONC WT 10.0MM/7.0MM 2-1/4CR F22M	4	4	
1380			RED CONC	5407238	3	3	REDUCER CONC WT 10.0MM/8.5MM 2-1/4CR F22M	4	4	
1390			RED CONC	5407207	10	10	REDUCER CONC WT 22.0MM/19.0MM 2-1/4CR F22M	12	12	
1400			RED CONC	5407531	10	10	REDUCER CONC WT 24.0MM/19.0MM 2-1/4CR F22M	14	14	
1410			RED CONC	5407532	12	12	REDUCER CONC WT 24.0MM/22.0MM 2-1/4CR F22M	14	14	
1420			RED CONC	5407533	10	10	REDUCER CONC WT 27.0MM/19.0MM 2-1/4CR F22M	16	16	
1430			RED CONC	5407534	12	12	REDUCER CONC WT 27.0MM/22.0MM 2-1/4CR F22M	16	16	
1440			RED CONC	5407266	16	16	REDUCER CONC WT 30.0MM/ 27.0MM 2-1/4CR F22M	18	18	
1450			RED CONC	5407264	14	14	REDUCER CONC WT 33.0MM/ 24.0MM 2-1/4CR F22M	20	20	
1460			RED CONC	5407263	16	16	REDUCER CONC WT 33.0MM/ 27.0MM 2-1/4CR F22M	20	20	
1470			RED CONC	5407267	18	18	REDUCER CONC WT 33.0MM/ 30.0MM 2-1/4CR F22M	20	20	
1480			RED CONC	5407273	16	16	REDUCER CONC WT 38.5MM/ 27.0MM 2-1/4CR F22M	24	24	
1490			RED CONC	5407276	18	18	REDUCER CONC WT 38.5MM/ 30.0MM 2-1/4CR F22M	24	24	
1500			RED CONC	5407271	20	20	REDUCER CONC WT 38.5MM/ 33.0MM 2-1/4CR F22M	24	24	
1510			RED CONC	5407260	24	24	REDUCER CONC WT 47.5MM/ 38.5MM 2-1/4CR F22M	30	30	
1520			RED ECC	5392882	0.75	1	REDUCER ECC SCH XXS LTCS A420 WPL6	1	1.5	
1530			RED ECC	5407508	0.75	1.5	REDUCER ECC WT 7.0MM / XXS 2-1/4CR F22M	2	2	
1540			RED ECC	5407509	0.75	1.5	REDUCER ECC WT 8.5MM / XXS 2-1/4CR F22M	3	3	
1550			RED ECC	5407510	1.5	1.5	REDUCER ECC WT 10.0MM / XXS 2-1/4CR F22M	4	4	
1560			RED ECC	5407511	2	2	REDUCER ECC WT 8.5MM / 7.0MM 2-1/4CR F22M	3	3	
1570			RED ECC	5407350	2	2	REDUCER ECC 10.0MM / 7.0MM 2-1/4CR F22M	4	4	
1580			RED ECC	5407351	3	3	REDUCER ECC 10.0MM / 8.5MM 2-1/4CR F22M	4	4	
1590			RED ECC	5407512	10	10	REDUCER ECC WT 22.0MM / 19.0MM 2-1/4CR F22M	12	12	
1600			RED ECC	5407513	10	10	REDUCER ECC WT 24.0MM / 19.0MM 2-1/4CR F22M	14	14	
1610			RED ECC	5407514	12	12	REDUCER ECC WT 24.0MM / 22.0MM 2-1/4CR F22M	14	14	
1620			RED ECC	5407340	10	10	REDUCER ECC WT 27.0MM/ 19.0MM 2-1/4CR F22M	16	16	
1630			RED ECC	5407535	12	12	REDUCER ECC WT 27.0MM / 22.0MM 2-1/4CR F22M	16	16	
1640			RED ECC	5407326	16	16	REDUCER ECC WT 30.0MM/ 27.0MM 2-1/4CR F22M	18	18	
1650			RED ECC	5407324	14	14	REDUCER ECC WT 33.0MM/ 24.0MM 2-1/4CR F22M	20	20	
1660			RED ECC	5407323	16	16	REDUCER ECC WT 33.0MM/ 27.0MM 2-1/4CR F22M	20	20	
1670			RED ECC	5407327	18	18	REDUCER ECC WT 33.0MM/ 30.0MM 2-1/4CR F22M	20	20	
1680			RED ECC	5407333	16	16	REDUCER ECC WT 38.5MM/ 27.0MM 2-1/4CR F22M	24	24	
1690			RED ECC	5407336	18	18	REDUCER ECC WT 38.5MM/ 30.0MM 2-1/4CR F22M	24	24	
1700			RED ECC	5407331	20	20	REDUCER ECC WT 38.5MM/ 33.0MM 2-1/4CR F22M	24	24	
1710			RED ECC	5407320	24	24	REDUCER ECC WT 47.5MM/ 38.5MM 2-1/4CR F22M	30	30	
1720							*****EXTRUDED BRANCHES*****			
1720										
1730	U04		EXTR PIPING	5506805E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 8.5MM x XXS - 2-1/4CR F22M	3	3	
1740	U04		EXTR PIPING	5506806E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 10.0MM x XXS - 2-1/4CR F22M	4	4	
1750	U04		EXTR PIPING	5506810E	2	2	NO MTO-EXTRUDED BRANCH - 10.0MM x 7.0MM - 2-1/4CR F22M	4	4	
1760	U04		EXTR PIPING	5506811E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 19.0MM x XXS - 2-1/4CR F22M	10	10	
1770	U04		EXTR PIPING	5506812E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 24.0MM x XXS - 2-1/4CR F22M	14	14	
1780	U04		EXTR PIPING	5506814E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 33.0MM x XXS - 2-1/4CR F22M	20	20	
1790	U04		EXTR PIPING	5506928E	3	3	NO MTO-EXTRUDED BRANCH - 33.0MM x 8.5MM - 2-1/4CR F22M	20	20	
1800	U04		EXTR PIPING	5506821E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 38.5MM x XXS - 2-1/4CR F22M	24	24	
1810	U04		EXTR PIPING	5506826E	0.75	1.5	NO MTO-EXTRUDED BRANCH - 47.5MM x XXS - 2-1/4CR F22M	30	30	
1820	U04		EXTR PIPING	5506875E	2	2	NO MTO-EXTRUDED BRANCH - 13.0MM x 7.0MM - 2-1/4CR F22M	6	6	
1830	U04		EXTR PIPING	5506824E	2	2	NO MTO-EXTRUDED BRANCH - 38.5MM x 7.0MM - 2-1/4CR F22M	24	24	
1840	U04		EXTR PIPING	5506825E	2	2	NO MTO-EXTRUDED BRANCH - 47.5MM x 7.0MM - 2-1/4CR F22M	30	30	
1850	U04		EXTR PIPING	5506831E	3	3	NO MTO-EXTRUDED BRANCH - 38.5MM x 8.5MM - 2-1/4CR F22M	24	24	
1860	U04		EXTR PIPING	5506832E	3	3	NO MTO-EXTRUDED BRANCH - 47.5MM x 8.5MM - 2-1/4CR F22M	30	30	
1870										
1870		816					***** FLANGES *****			

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - NONE	CORR. ALLOW. 3 MM	SOUR SERVICE NACE MR0175/ISO 15156 DESIGN: ASME B16.5 MG 1.10 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: 2-1/4CR F22 MODIFIED FLANGE RATING: 1500 RF/CF TEMPERATURE RANGE: -40 / 155 °C	12/16/2024	BY OZAC	U04	Page 3 of 5	1500F22

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1880										
1880	U04	FA8	FLANGE	5599704	0.5	0.5	FLG SCRD (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	0.75	
1890										
1890	U04		FLG ADAPTER	5851205	0.75	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1900										
1900	U04	UDA	FLANGE	5557505	0.75	1.5	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
1910	U04	UDE	FLANGE	5563338	0.5	1.5	FLG WN 1500# RF 2-1/4CR F22M SCH XXS	0	0	
1920			FLANGE	5563350	2	2	FLG WN 1500# RF 2-1/4CR F22M WT 7.0MM	0	0	
1930			FLANGE	5563351	3	3	FLG WN 1500# RF 2-1/4CR F22M WT 8.5MM	0	0	
1940			FLANGE	5563352	4	4	FLG WN 1500# RF 2-1/4CR F22M WT 10.0MM	0	0	
1950			FLANGE	5563353	6	6	FLG WN 1500# RF 2-1/4CR F22M WT 13.0MM	0	0	
1960			FLANGE	5563354	8	8	FLG WN 1500# RF 2-1/4CR F22M WT 16.0MM	0	0	
1970			FLANGE	5563355	10	10	FLG WN 1500# RF 2-1/4CR F22M WT 19.0MM	0	0	
1980			FLANGE	5563356	12	12	FLG WN 1500# RF 2-1/4CR F22M WT 22.0MM	0	0	
1990			FLANGE	5563380	14	14	FLG WN 1500# RF 2-1/4CR F22M WT 24.0MM	0	0	
2000			FLANGE	5563381	16	16	FLG WN 1500# RF 2-1/4CR F22M WT 27.0MM	0	0	
2010			FLANGE	5563382	18	18	FLG WN 1500# RF 2-1/4CR F22M WT 30.0MM	0	0	
2020			FLANGE	5563383	20	20	FLG WN 1500# RF 2-1/4CR F22M WT 33.0MM	0	0	
2030			FLANGE	5563385	24	24	FLG WN 1500# RF 2-1/4CR F22M WT 38.5MM	0	0	
2040			FLANGE	5563429	30	30	FLG WN 1500# RF 2-1/4CR F22M WT 47.5MM MANUFACTURERS STANDARD	0	0	
2050	U04	UDE	FLANGE	5563339	1	1	FLG WN 2500# RTJ 2-1/4CR F22M SCH XXS	0	0	
2060	U04	817	FLANGE	5563363	3	3	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 8.5MM	0	0	
2070	U04	817	FLANGE	5563364	4	4	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 10.0MM	0	0	
2080	U04	817	FLANGE	5563365	6	6	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 13.0MM	0	0	
2090	U04	817	FLANGE	5563366	8	8	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 16.0MM	0	0	
2100	U04	817	FLANGE	5563367	10	10	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 19.0MM	0	0	
2110	U04	817	FLANGE	5563368	12	12	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 22.0MM	0	0	
2120	U04	817	FLANGE	5619832	14	14	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 24.0MM	0	0	
2130	U04	817	FLANGE	5619833	16	16	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 27.0MM	0	0	
2140	U04	817	FLANGE	5619834	18	18	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 30.0MM	0	0	
2150	U04	817	FLANGE	5619835	20	20	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 33.0MM	0	0	
2160	U04	817	FLANGE	5619836	24	24	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 38.5MM	0	0	
2170	U04	817	FLANGE	5619837	30	30	FLG COMPACT 1500# RJ 2-1/4CR F22M WT 47.5MM	0	0	
2180			FLANGE	5563311	10	10	FLG WN 2500# RTJ 2-1/4CR F22M WT 19.0MM	0	0	
2190	U04		BLIND	5615616	0.5	1.5	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
2200			BLIND	5407447	2	2	FLG BLIND 1500# RF 2-1/4CR F22M	0	0	
2210			BLIND	5407447	3	24	FLG BLIND 1500# RF 2-1/4CR F22M	0	0	
2220			BLIND	5407496	30	30	FLG BLIND 1500# RF 2-1/4CR F22M MANUFACTURERS STANDARD	0	0	
2230	U04	817	BLIND	5619782	3	3	FLG COMPACT BLIND 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 3 INCH ONLY	0	0	
2240	U04	817	BLIND	5619781	4	30	FLG COMPACT BLIND 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2250							***** ORIFICE FLANGES *****			
2250										
2260										
2260			ORIFICE FLG	5407585	24	24	WN ORIF SET 1500# RF 2-1/4CR F22M WT 38.5MM	0	0	
2270										
2270		812					***** PIPE NIPPLES *****			
2280										
2280	U04		NIPPLE	5541181	0.5	1.5	NIPPLE SCH XXS 2-1/4CR F22M BBE 4" LG	0	0	
2290										
2290							***** GASKETS *****			
2300										
2300			GASKET	5672251	0.5	24	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
2310			GASKET	5674013	30	30	GASKET 1500# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
2320	U04	GA6	GASKET	5673021	2	3	GASKET 1500# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
2330	U04		GASKET	5674480	1	1	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2340			GASKET	5674480	10	10	GASKET 2500# RTJ SS OCTAGONAL RING	0	0	
2350	U04	817	GASKET	5673471	3	3	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500F22 FOR NPS 3 INCH ONLY	0	0	
2360	U04	817	GASKET	5673470	4	30	GASKET 1500# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2370	U04		GASKET	5674483	12	14	GASKET 10000 API 6A BX 158 11" NS 316SS RING	0	0	
2380	U04		GASKET	5674484	16	18	GASKET 10000 API 6A BX 159 13 5/8" NS 316SS RING	0	0	
2390										
2390		B1					***** BOLTING *****			
2400										
2400	U04		BOLTING	5675770	0.5	4	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2410	U04	BA1	BOLTING	5675770	6	14	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
2420	U04	BA5	BOLTING	5675778	6	99	STUD-BOLT C/W 2 COMPACT NUTS API-6A718	0	0	
2430		BA1	BOLTING	5675776	16	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
2440										
2440		SB6	BOLTING	5675675	12	30	JACK SCREW HEX HEAD A320 GR L7M	0	0	
2450										
2450							***** BLINDS & SPACERS *****			

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
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SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
2460										
2460	U04	817	SPEC BLIND	5749577	4	4	SPECTACLE BLIND COMPACT 1500# RJ 2-1/4 CR F22M PIPE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2470										
2470	U04	817	PADDLE	5750292	3	3	PADDLE BLANK COMPACT 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 3 INCH & BELOW	0	0	
2480	U04	817	PADDLE	5750293	3	3	PADDLE SPACER COMPACT 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 3 INCH & BELOW	0	0	
2490	U04	817	PADDLE	5750286	4	4	PADDLE SPACER COMPACT 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2500	U04	817	PADDLE	5750286	10	30	PADDLE SPACER COMPACT 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2510	U04	817	PADDLE	5750285	4	4	PADDLE BLANK COMPACT 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2520	U04	817	PADDLE	5750285	10	30	PADDLE BLANK COMPACT 1500# RJ 2-1/4CR F22M LINE CLASS 1500F22 FOR NPS 4 INCH & ABOVE	0	0	
2530										
2530			SPEC BLIND	5407647	2	2	SPECTACLE BLIND 1500# RF 2-1/4 CR F22M	0	0	
2540			SPEC BLIND	5407647	3	4	SPECTACLE BLIND 1500# RF 2-1/4 CR F22M	0	0	
2550										
2550			PADDLE	5407781	10	24	PADDLE SPACER 1500# RF 2-1/4CR F22M	0	0	
2560			PADDLE	5407881	10	24	PADDLE BLANK 1500# RF 2-1/4CR F22M	0	0	
2570			PADDLE	5407782	30	30	PADDLE SPACER 1500# RF 2-1/4CR F22M MANUFACTURERS STANDARD	0	0	
2580			PADDLE	5407882	30	30	PADDLE BLANK 1500# RF 2-1/4CR F22M MANUFACTURERS STANDARD	0	0	
2590										
2600										
2600			OLETS	5514136	0.75	1.5	NIPOFLANGE 1500# RF 33.0MM/XXS 2-1/4CR F22M	20	20	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
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